WCS 28
I want to donate my organs:
Brain death; Organ donation; Psychological issues

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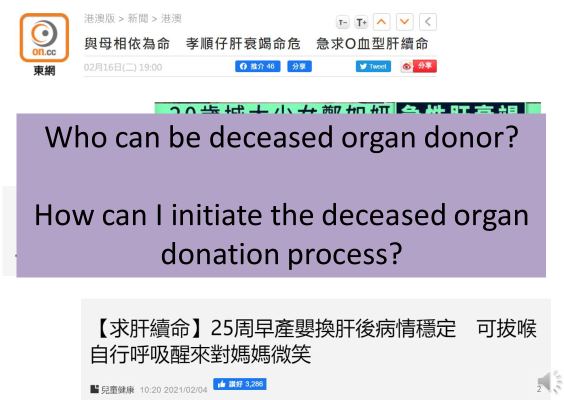

Who can be deceased organ donor?

How can I initiate the deceased organ donation process?

Deceased Organ Donation Process

Manage the 'potential donor'
-To optimize the function and viability of all transplantable organ

- Identified Potential Organ Donor
  Inform Organ Donation Coordinator
- Assessment of Potential Donor
  Confirm brain death
  - No contraindication for donation
  - Brain death confirm
- Discuss the options of organ/tissue donation and obtain family's consent *(Opt-in system)*
  - Family consent obtained
- Prepare donor for organ recovery

When shall we trigger referral for Deceased Donor organ donation?

- Clinical Referral trigger:

- **G CS<5**
- **I RREVERSIBLE BRAIN INJURY**
- **V ENTILATED**
- **E ND OF LIFE CARE**

GIVE Organ Donation Coordinator (ODC) a call to trigger the deceased organ donation process

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Deceased Organ Donation in Hong Kong

General criteria for deceased organ and tissue donation:

- No age limit in general
- Type of donors:
  - Brain death donors: donate both organs and tissue
    - Organ function temporarily maintained by ventilatory and drugs support
  - Cardiac death donors: donate tissue only, such as corneas, skin and bones
    - Organ donation after cardiac death is increasing around the world)
- Has adequate organ function
- No severe / systemic infection
- No HIV infection
  - Some countries do accept HIV +ve donor
- Cancers are contraindicated in organs, skin and bone donation, except primary brain tumours.
  - However, selected deceased with cancer may be considered cornea donation

Diagnosis of Brain Death

- Definition of Brain death
  - Irreversible cessation of brainstem function
  - Established by the documentation of
    - irreversible coma
    - irreversible loss of brain stem reflex responses and respiratory center function
    - Demonstration of the cessation of intracranial blood flow
- The diagnosis of brain death is made by the separate examination of *two doctors* who have no relations with organ transplant service to carry out *two separate tests*

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Diagnosis of Brain Death

- One need to exclude reversible cause of coma
  - χ Effect of sedative drugs and/or muscle relaxant
  - χ Hypothermia (i.e. body temperature<35C)
  - χ Metabolic and endocrine disturbance
    - There should be no profound abnormality of electrolyte, acid base balance, or blood glucose concentration
  - χ Arterial hypotension
- Neuromuscular function is intact

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Diagnosis of Brain Death

- Clinical Test of Brain Stem Function
  - Both pupil are fixed and not respond to light
  - Corneal reflex is absence
  - Vestibulo-ocular reflex is absence
    - No eye movement on cold caloric testing
  - No gag or cough reflex
  - No motor response in the cranial nerve distribution to stimulation to face, limbs or truck
  - Apnoea test
    - No respiratory movement with arterial PCO_2>8kPa and pH<7.3

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Diagnosis of Brain Death

- Confirmatory test would be needed if:
  - No clear cause of coma
  - Possible metabolic effect or drug effects
  - Cranial nerves cannot be adequately tested
  - Cervical vertebral or cord injury
  - Cardiovascular instability precluding apnoea test
  - Severe hypoxaemic respiratory failure precluding apnoea test
- Confirmatory tests: Four vessel radio-contrast angiography with digital subtraction
  - to demonstrate absence of intracranial blood flow

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Prepare donor for organ recovery

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Opt-In vs Opt-Out

- Opt-In *(adopted in Hong Kong now)*
  - People must actively sign up to a register to donate their organs after death *(Centralised Organ Donation Register, CODR: codr.gov.hk)*
  - If a person doesn't register, the family will make to decision at the time of death

- Opt-out
  - Organ donation will occur automatically unless a specific request is made before death for organ not to be taken

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Dos' and Don'ts' when communicate with family of potential deceased donor

Dos'

- Do identify and make referral to Organ Donation Coordinator when there is a potential organ donor
- Do maintain vital sign and haemodynamic stability of the potential organ donor through the process
- Do respect family regarding the option of organ donation
- Do assure confidentiality of donor's record

Don'ts'

- Don't discuss choice of organ donation before brain death certification
- Don't screen cases for the transplant team
- Don't take away the chance of 'Those' willing to donate and 'Those' waiting for a transplant

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Proper Management of Imminent Brain Death Patients is Necessary……

- Stabilize the patient
  - Facilitate brain death examination
- Manage the "potential donor"
  - To optimize the function and viability of all transplantable organs

What to expect in an Imminent Brain Death Patient?

Loss of brain stem function results in systemic physiologic instability……

- Unstable blood pressure
  - hyper- and hypotension
- Cardiac arrhythmia
- Loss of temperature regulation
- Hormonal imbalance
  - Cranial diabetes insipidus
- Loss of respiratory function
- Electrolyte disturbance

Loss of brain stem function results in systemic physiologic instability……

- Unstable blood pressure
  - hyper- and hypotension

Hypertensive phase

- ↑ICP: ↑arterial blood pressure to ensure adequate cerebral perfusion pressure (Cushing's reflex)
- Autonomic/ Sympathetic storm
  - Release of catecholamines and results in hyperdynamic states
  - Tachycardia, peripheral vasoconstriction and hypertension

Hypotensive phase

- Sympathetic outflow loss, catecholamine depletion, myocardial dysfunction, intense peripheral vasodilatation, hypovolemia, electrolyte disturbance and endocrine changes

Loss of brain stem function results in systemic physiologic instability……

- Unstable blood pressure
  - hyper- and hypotension
- Cardiac arrhythmia
- Lo
- H
- Lo
- El

- Atrial and Ventricular dysrhythmias, conduction blocks
- Due to electrolyte and acid-base disturbances, hypothermia, decreased myocardial contractility, inotrope use and increased ICP

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Due to loss of hypothalamic temperature regulation

Hypothermia can induce coagulopathy, haemolysis and leftward shift of oxyhaemoglobin dissociation curve

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- Due to posterior hypothalamic-pitutary deficiency
- Causes polyuria, hypovolaemia, hypotension and hypovolaemic hypernatraemia

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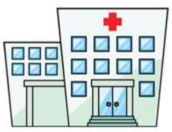
Management of Imminent Brain Death Patients



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Targets

- Maintain arterial SBP 100-140mmHg/ MAP ~ 65mmHg
- Maintain AR 60-120 bpm
- Maintain hourly urine output ~ 100ml
- Maintain SaO2 >90%
- Maintain body temperature >36ºC
- Maintain normal ranges for serum glucose, potassium, calcium, phosphate, magnesium

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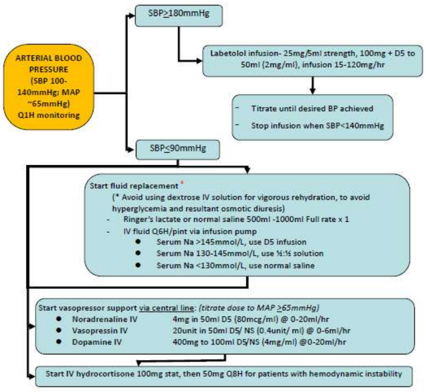
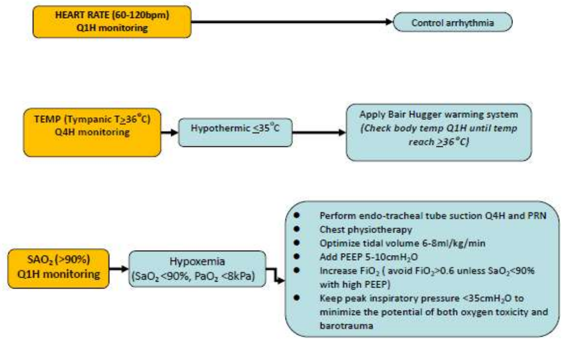


Adopted from **CCOT Guideline for Support of Imminent Brain Death Patient in Medical Ward**



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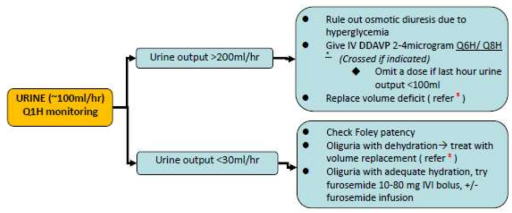

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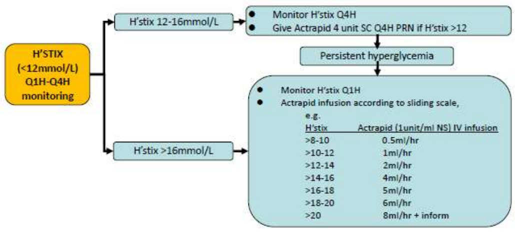

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Take home message

- Early referral to organ donation coordinator for assessment
  - G CS<5
  - I RREVERSIBLE BRAIN INJURY
  - V ENTILATED
  - E ND OF LIFE CARE
- **DO NOT** discuss choice of organ donation with family before brain death certification
- Currently only brain death donors are accepted for deceased organ donation in Hong Kong
  - The diagnosis of brain death is made by the separate examination of *two doctors* who have no relations with organ transplant service to carry out *two separate tests*
  - Systemic physiological instability is expected in imminent brain death patients
  - Management of imminent brain death needs clinical expertise, vigilance and ability to address multiple clinical problems effectively and simultaneously
  - Collaboration with Organ Donation Coordinators and Critical Care staff can further improve potential donor management

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